

MS2 Bacteriophages and Antibiotics on E. coli In Computational Simulations

AP Research

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Introduction

Antibiotic resistance is an issue in not just developing countries, but also developed countries. *Escherichia coli* in particular develops antibiotic resistance rapidly, most prevalently through horizontal gene transfer, also known as conjugation (Nasrollahian et al., 2024). The extensive overuse of over-the-counter antibiotics within the world has contributed to the large increase in antibiotic resistance, done often without veterinarian approval (Adhikari, Lippmann, Parajuli, 2024). This is done through many mechanisms, such as efflux pumps, extended spectrum beta-lactamases, and carbapenemases. The rising trend of *E. coli* infections within leafy greens poses a legitimate problem, as *E. coli* is the most common pathogen associated with leafy green infections, causing diarrhea, hemorrhagic colitis, and other infections (Luna-Guevara et al., 2019). In 2018-2020, multiple infections of OH157 *E. coli* yielded patterns that led the United States Food and Drug Administration to find a recurring pattern, that issues with adjacent activities like cattle feces, migrating birds, and other activities often resulted in cross-contamination, as well as the geographical location of the farms (Lacombe et al., 2022). Interestingly, the majority of *E. coli* outbreaks were a result of *E. coli* OH157 producing a potent toxin, Shiga Toxin 2, as well as having an extraordinarily small infectious dose, 1-100 viable cells, and often causing complications during treatment (Centers for Disease Control, 2026; LeBlanc, 2003). Shiga toxin 2 expressing bacteria also proliferates within many vectors, including milk (cow), beef, vegetables, flour, pork, nuts, sausage, chicken, and most animal-based food products (Wang et al., 2024). Therefore, a solution must be created to remove a large portion of an *E. coli* population to the extent where rapid reproduction of *E. coli* cells are hindered or removed, and a part of this solution must be to address antibiotic resistance. With the increase of testing on humans, using bacteriophages, many clinical trials have been successful, though the number of

formal clinical trials remains closed from strict human testing limitations, including the mass of research in Russian and Polish journals that are difficult to access (Romero-Calle et al., 2019). One field that has been explored for this solution includes the usage of bacteriophages on plant diseases, which was not recommended due to ultraviolet light, heat, and pH concerns (Jones et al., 2012). However, a solution to this issue is the application of bacteriophages on the surface of plants is for the prevention of phytopathogenic bacteria, through a spray or surface compound (Korniienko et al., 2022). Therefore, applications of bacteriophage therapies onto plant surfaces through a spray or surface compound presents a convincing solution to the wastewater and splash-based contamination methods of *E. coli* in lettuce, bean sprouts, cantaloupes, and many other produce (Hilborn et al., 1999; Ackers et al., 1998). In more recent years, the majority of these issues were found in premade salads and Romaine Lettuce as a result of bacteria infiltrating cuts and bruises within the lettuce, even more so while washing lettuce plants (Uhlig et al., 2017). Specifically, Shiga toxin-producing *E. coli*, Norovirus, *Campylobacter jejuni*, and nontyphoidal *Salmonella* make up the highest number of illnesses and costs in leafy green outbreaks (Yang et al., 2024). Furthermore, Yang found that 75.7% of leafy green foodborne illnesses are a result of Romaine lettuce being infected. Notably, the location of various farms within Yuma County played large roles within the infection of plants (United States Food and Drug Administration, 2021). A solution including facets for antibiotic resistance, cost, prevention of cross-transmission, and possible sprays for bacteriophage applications onto plants can resolve an issue of plant-based transmissions of viral bacteria, contribute to the gap of knowledge on antibiotic-bacteriophage synergies, and contribute to the application of bacteriophages to target non-phytogenic bacteria on plants.

Literature Review

F plasmid Mechanics and Architecture

F plasmids are 100-kbp circular conjugative plasmids and are vectors for horizontal gene transfer and gene recombination. Currently, more than 200 F-like plasmids exist, and in the 60 kbp backbone, DNA transfer genes are the most conserved apart and occupy the majority of the backbone (Koraimann, 2018). F-like plasmids seem to be restricted to closely related enterobacterial genera, meaning the F plasmid is an attractive target for a generalized bacteriophage target. Interestingly, F plasmids are often found in commensal and pathogenic *E. coli* (Koraimann, 2018). The original F plasmid is constitutively derepressed because the *finO* gene is disrupted by an insertion sequence IS3 (Cheah et al., 1987). The *finO* gene encodes a 22 kDa fertility inhibition protein (FinO) in *Escherichia coli* and other Gram-negative bacteria, aiding the antisense RNA FinP in binding and silencing the *traJ* gene, which prevents the transfer of F-like plasmids between bacteria (Van Biesen et al., 1994; Sandercock et al., 1994). This means bacteria with a F-like plasmid will perpetually have an extended F pili, meaning MS2 bacteriophage can attack said target through the pili at any time, contributing to the possible effectiveness of MS2 bacteriophages as a vector for antimicrobial effects. Towards the operation of conjugation, F plasmids are true type 4 secretion systems, which is a nanomachine the F plasmid uses to move DNA from a donor cell to a recipient cell (Lawley et al., 2003). The T4S system contains a T4P pilus, which is what MS2 bacteriophages attach to during lysis, and a T4C complex that crosses the periplasm. Later, this is significant towards stoichiometrical modeling of MS2 bacteriophage and ampicillin interactions because this demonstrates a possible necessity for port and metabolic interactions for composite models. For the introduction of T4P and T4S machinery, the gene product “*slt*” locally degrades the peptidoglycan layer in a recipient cell,

allowing the assembly of the T4S complex (Bayer et al., 1995). F pili differs from other bacteria because it is flexible, retracts to bring two cells together, is plasmid encoded versus chromosome encoded, and is mostly used for DNA transfer versus adhesion (Wang et al., 2009).

Persistence, Themes, and Benefits of F Plasmid Targeting

F plasmids are some of the first “superspreader mutations,” dramatically enhancing the conjugational efficiency of conjugative plasmids that belong to diverse incompatibility groups (Virolle et al., 2020). Antibiotic resistance genes are located mainly in an F plasmid; 66% of acquired resistance genes are located in F replicons within *E. coli* (Stephans et al., 2020). In fact, F plasmids have enhanced transfer capacities compared to IncF plasmids, R100, R6-5, and R1 plasmids branching from the F-class plasmid, using the active FinOP regulatory system (Yoshioka et al., 1987). Lederburg et al.’s 1952 paper on sex compatibility, the discovery of F plasmids and the mechanics of F plasmid spread demonstrates the earliest mention of F plasmid discovery, most notably that the presence of a (then deemed) “F factor” (F⁺ vs F⁻) would mean most bacteria that engaged with this F factor would gain the F⁺ status (Lederburg et al., 1952). >50% of all plasmids can be transferred by conjugation, and most evolutionary models assuming that plasmids must be mobile because plasmid carriage incurs a fitness cost, and the general heterogeneity of male-specific plasmids create the circumstances where male-specific plasmids and mobile plasmids are evolutionarily beneficial despite considerable reproduction drawbacks and fitness costs (Smille et al., 2010; Dionisio, 2005). The cost of antibiotic resistance over time can be substantially distinguished or eliminated by evolution over short periods of time, meaning shortly suspending the use of antibiotics is not an actually viable solution towards antibiotic resistance (Lenski, 1998). However, the usage of MS2 bacteriophages and ampicillin as phage-antibiotic synergetic combinations towards the eradication of bacteria is significant

because it presents the capacity to apply two evolutionary pressures: the pressure to lose the F plasmid or be lysed, and the pressure to keep the F plasmid or be lysed by ampicillin. The usage of conjugation is sufficiently fast to exhibit persistence and is generally applicable to communities containing multiple plasmids and multiple populations (Lopatkin et al., 2017). The speed of conjugation and lysis rate competes with each other, creating a unique circumstance where lysis could lyse all cells containing the F plasmid before the F plasmid persists, a condition that will be monitored in the experiment. After this study on persistence and reversal of plasmid-mediated antibiotic resistance, Lopatkin et al. found that there is a necessity to inhibit the conjugative process for effective resistance reversal. Penttinen et al.'s paper on indirect selection against antibiotic resistance through plasmid-dependent bacteriophages further shows that even if phage-mediated selection is not an effect of immediate eradication, phages can still interfere with the “natural order of things in the microbial realm where evolution towards resistance becomes reality” (Penttinen et al., 2021). One such possibility is the MS2 bacteriophage, MS2 standing for male-specific, meaning this bacteriophage specifically selects and proceeds through the lytic cycle for bacteria exhibiting the F plasmid's pili (Spankie et al., 2020). MS2 bacteriophages infect the bacterium *E. coli* through the F-pilis on *E. coli* bacteriophages, or any bacterium with the F⁺ factor (the F plasmid), by uncoating at any point along the *E. Coli* pilis at a distance from the cell body, allowing for the lysis of bacteria (Paine et al., 2025; Cock et al., 2010). After the initial extracellular uncoating, the MS2 bacteriophage uncoats at the cell surface after F-pilis retraction and replicates within the bacteria, ultimately lysing the cell from within. Interestingly, Cock et al. found that MS2 bacteriophages can infect *S. aureus* as well as F⁺ *E. coli*. However, it must be noted that application of bacteriophages to antibiotic resistant bacteria selects for antibiotic sensitive bacteria cells that lose their plasmids

spontaneously and are bacteriophage resistant and thus will end up dominating the bacteria environment. Because the usage of MS2 bacteriophages select for antibiotic sensitive bacteria cells, the literature demonstrates that there is a large possibility for ampicillin - antibiotic synergisms. Bacteriophage resistance can appear in many capacities, mostly emerging in phage absorption prevention, preventing DNA integration by the Superinfection exclusion system, which is when a single protein blocks subsequent infections after an initial infection by blocking infections on various points, degrading phage DNA through restriction-modification systems and clustered regularly interspaced short palindromic repeats (CRISPR), which cut DNA through restriction enzymes and RNA molecules to recognize and target specific DNA sequences (Drulis-Kawa et. al, 2012; Moghadam et. al, 2020). Bacteria lysins bypass these defences by having a high specificity, cleaving highly conserved structures on bacterium walls, and because lysins are applied externally versus internally, bypassing many traditional resistance methods like active reflux and reduced membrane permeability (Moghadam et. al, 2020; Wei, Yang, Yu, 2014). Notably, the loss of a receptor(s) on a bacterium are only within inverse gene for gene coevolution patterns, whereas gene for gene coevolutions involve gaining the ability to recognize a portion of a bacteriophage, such as the flagella, and the loss of said portion in the bacteriophage (Burmeister et. al, 2021).

Bacteriophage Synergy

Interestingly, results demonstrated by Cvetanovic et al. showed how *Microvidae* and MS2 bacteriophages can still be successful if used in synergy with existing antibiotic treatments (Cvetanovic et al., 2024). Phages can evolve or co-evolve and consequently restrict the evolution of resistance mechanism in bacteria, synergistic bacterial activity of an isolated phage and ampicillin against urinary tract infecting *Escherichia coli* O157 found that sub-inhibitory

concentrations of ampicillin in mixture with the isolated natural phage was a good alternative to using high doses of antibiotics (Moradpour et al., 2020). This presents a convincing alternative or supplement to traditional usages of antibiotics in agriculture. Lytic bacteriophages in combination with antibiotics show phage-antibiotic synergy in a separate study, where the timing sequence of administering phages and antibiotics can impact the killing effects on different bacterial strains (Liu et al., 2006). The study found that ampicillin had an affinity for PBP2s and PBP3s (penicillin-binding proteins that help construct the cell wall of gram-negative bacteria. In *Salomonella Typhimurium* phage-resistant bacteria also showed signs of reduced virulence and antibiotic resistance, and there was an inverse effect in reference to bacteriophage and antibiotic sensitivities in the targeted bacteria (Laure et al., 2022). This suggests that MS2 bacteriophages can not only evolutionarily select for reduced virulence, the loss of antibiotic resistance overall but also antibiotics can also help contribute to an emphasis of these effects. Phage antibiotic synergy overall is cheaper and more resource efficient in comparison to phage cocktails (Biswas et al., 2024). Interestingly, though overall 2x and 1x minimum inhibitory concentrations (MIC) strains showed no significant difference compared to individual or combination treatments, but at 0.5 MIC values, there was clear PAS (Rastegar et al., 2024). Overall, MS2 bacteriophages are good solutions for the treatment of *E. coli* and other bacteria with the F plasmid gene or F-like plasmid genes because they select against having resistance. This selection allows for a cheaper solution to bacteriophage resistances considering existing materials, follows literature demonstrating that lytic bacteriophages and antibiotics already have synergy, and shows great potential for synergy with the evolutionary forces that ampicillin natively provides.

Digital Modeling and Stoichiometric Constraints

Vivarium is an open source software framework developed by the Covert Lab at Stanford which possesses the ability to build composite models combining many useful modeling frameworks: agent-based models, ordinary differential equations, and stoichiometric reaction systems (Agmon et al., 2022). Vivarium was designed with the goal of integrated whole-cell modeling, but enables interfaces between various ranges of different models in a unified simulation. The nested hierarchies and scalable representation Vivarium was designed with allows for rigid-body physics simulators which can be used to model MS2 bacteriophage particles, ampicillin particles, the movement of bacteria cells, and the interaction of various cellular elements and external environments through modules like bioscrape, genEcoli, and multibody, which can be found on the vivarium github (Vivarium Collective, n.d.). Around that time, Vivarium was used to successfully model cell-cell protein interactions coupled with external ampicillin and heterogeneous gene interactions (Skalnik et al., 2021). The flexibility of using Vivarium was that models of antibiotic resistance could be added, and the colony's reaction could be more diversified and simulated. An Agent-Based Model (ABM) is superior to ordinary differential equations and simplified population dynamics through derivative equations because of Aljuhani et al.'s paper on quantification of ampicillin diffusions, being at a rate of 3×10^{-6} to 5×10^{-6} centimeters²/second, yet Baltus et al. showed that the free diffusivity constant of the icosahedral MS2 bacteriophage was 3.3×10^{-8} to 1×10^{-7} cm²/s (Aljuhani et al., 2024; Baltus et al., 2016). These particles move at different rates and individually simulating particles is advantageous because traditional differential equations are deterministic and average out the behavior of a system, often assuming perfectly mixed solutions and infinite populations, but Gillespie's equation captures extrinsic noise and the discrete, random nature of particle collisions (Gillespie, 1977). Despite the age of Gillespie's algorithm, it is still one of two fundamental

approaches to modeling, and since the simplistic nature and modulated nature of Vivarium's multi-CPU based framework allows for fast loading times, the remarked drawback of loading times is not an issue (Albert, 2016; Agmon et al., 2022). Equations treat all *E. coli* in one generalized state; ABMs are superior because it allows for the state of each individual cell, such as F plasmid status, which is the goal of the experiment anyways. Elowitz et al. shows that cells show substantial phenotypical variation, and that intrinsic and extrinsic noise should be factored in (Elowitz et al., 2002). In a separate paper, Gillespie describes the change of chemical populations being by discrete integer amounts, a "random-walk process" because Ordinary Differential Equations (ODE) describe the same result every time, yet Gillespie's Monte Carlo Algorithm allows for the probability of an outcome and inherent fluctuations (Gillespie, 1977). Though Monte-Carlo equations are good for multiple model evaluations using random or pseudo-random numbers (e.g. numbers gathered from a standard deviation range centered at a given constant, usually 1) multiplied to a set array or algorithm, Latin Hypercube Modeling (LHM), introduced by McKay et al. in 1979, allows for an unbiased estimate of the original model's results, requiring fewer samples than simple random sampling (Marino et al., 2009; McKay et al., 1979). To make sure the synergy evaluations and results were robust, LHM is used for global sensitivity analysis to ensure a more uniform coverage of multidimensional input spaces. Interestingly, Krysiak-Baltyn et al. describe slight differences in absorption, lysis, etc... rates often causing heavy oscillations in simulations and often displaying unreliable results, meaning the global sensitivity of parameters should be tested (Krysiak-Baltyn et al., 2016). There is a distinct description of a lack of "quantitative comparison" and that more emphasis should be placed upon standardized sensitivity tests like Latin Hypercube Modeling. Moreover, Krysiak-Baltyn et al. describe a large set of assumptions unfounded from *in vitro* testing, such as

the assumption of bacteria remaining in log-phase of growth, highly non-linear models being represented linearly, heavy oscillations where stable states are, with many issues isolated in models that are not strictly even for bacterial growth. Overall, the gap of literature shows an unexplored field: quantified synergies between MS2 / male specific bacteriophages and Ampicillin within a stochastic, agent-based framework that accounts for both intrinsic biological noise and multi-system interactions. Furthermore, the evolutionary pressures for the loss of antibiotic resistance in the form of the loss of F plasmids, a pressure to keep F plasmids, and overall fitness costs of F plasmids will be explored and answered.

Methods

Bias and Ethics

Bias within this study resembles computational, coding, and analysis bias. In order to account for these issues, data analysis must strictly be from existing data generated from the code, and peer-reviewed data analysis sources. Slinker promotes the usage of a “factorial ANOVA (analysis of variance)” as a method to distinguish between a major effect and synergism, assuming that the two treatments given will be an interacting force, focusing on an “interaction term” (Slinker, 1998). Slinker’s procedure helps prevent confirmation bias because it gives a value separating synergism and effect versus a dominant phage energy. The Bliss Independence model was then used for the assumption that two interactions were, as named, independent (Foucquier et al., 2015). Bliss states that the Bliss Independence model assumes that “drug [any two antibiotics, in a modern interpretation, this includes viruses that target bacteria] effects are outcomes of probabilistic processes” (Bliss, 1939). These two models are meant to confirm or reject any synergy, and leave no room for interpretive bias. Coding bias is addressed through strictly utilizing equations, Vivarium processes, and coding strictly through the

literature. Ethical issues include mostly the usage of AI, which is circumvented by simply not using AI and using the literature to write code because previous researchers have demonstrated the appropriate equations and basic research can allow a researcher to learn python to utilize the correct functions to run these equations. There is likely no superbug capacity within this research considering the lack of possible bacteriophage resistance. In vitro studies must occur before any information from this paper can be applied to clinical trials, which likely would not occur considering the results of the paper anyways. This solution is cheap enough to be applied to the medical industry without any significant cost, unless a company chooses otherwise, and therefore can be applied to a larger population, requiring simply one antibiotic and one bacteriophage.

Variables

The variables are natural growth, growth with ampicillin, with and without F+ factors, and 3 ampicillin concentrations based off of an "original" ampicillin value obtained from the literature (0.5x, 1x, and 1.5x values) were tested in combination with MS2 bacteriophages. This allows for a basis of comparison for the factorial ANOVA and allows for early visual analysis of data based on previous literature to make sure the code is functional.

Table 1

Control Types

Control type	Ampicillin values	Phage values	Plasmid Status
Growth Control	0	0	True
F plasmid Control	500	0	False

Ampicillin Control	500	0	True
Phage Control (Phage, no F gene)	0	50	False
Phage Control (Phage, F+)	0	50	True
Treatment: Low Amp (143.1 uM)	143.098365	50	True
Treatment: Mid Amp (286.2 uM)	286.19673	50	True
Treatment: High Amp (572.4 uM)	572.39346	50	True

Hypothesis

Null

MS2 bacteriophages and ampicillin will not have any effect upon each other and will not contribute or remove from the overall results.

Experimental

MS2 bacteriophages and ampicillin will show signs of synergism through the loss of F plasmids and therefore lysis by ampicillin.

Alternate

Ampicillin will have a negative effect on MS2 bacteriophages and will reduce the rate of lysis.

Code and Interaction Logic

Marino et al. argued that for a simulation to have any “predictive power” it must be based on biologically plausible ranges (Marino et al., 2009). Therefore, a key selection of values were chosen directly from the literature.

Table 2

Values From the Literature

Statistic and Source	Value
Aqueous diffusion of small-molecule antibiotics [ampicillin] (Alqahtani et al., 2024)	$3 * 10^{-6}$ to $5 * 10^{-6}$ centimeters ² / second
The free diffusivity coefficient of the icosahedral MS2 bacteriophage (Baltus et al., 2016)	$3.3 * 10^{-8}$ to $1 * 10^{-7}$ centimeters ² /second
The exponential decay constant of the MS2 bacteriophage in liquid media [a converted value] (Yates et al., 1985)	$1.5 * 10^{-4}$ minute ⁻¹
The MS2 bacteriophage interaction and infection rate [a converted value]	$1 * 10^{-9}$ mL * min ⁻¹

(De Paepe et al., 2006)

Metabolic burden parameters (plasmid

2% - 5% compared to F- cells

bearing F+ cells)

(Händel et al., 2013)

Michaelis-Menten kinetics of F-plasmid
mediated Ampicillin degradation (general
Ampicillin resistance)

$V_{\max} = 2.7 * 10^4 \text{ min}^{-1}$, $K_m = 160 \text{ um}$

(Lundbäck et al., 1974)

Spontaneous segregational loss possibility

10^{-4} per cell generation (chance)

(Tucker et al., 1984)

Doubling rates (laboratory conditions)

20 minutes until doubling

(Sezonov et al., 2007)

Towards the framework chosen, Vivarium was used (Agmon et al., 2022). The specific modules chosen were multibody, bioscrape, and genEcoli for their spatial dynamics, stochastic simulation processes, and metabolic fluxes, as well as general interactions (Vivarium Collective, n.d.) First, imports and general setup were performed in the Jupyter notebook for superior organization, though it ended up being in one cell only because it was easier to do so. The imported modules and processes were pandas, numpy, matplotlib, qmc (Quasi-Monte-Carlo) and norm for LHM and Monte-Carlo from scipy, process, composer, and engine from Vivarium individually. A class was created “GenEcoliProcess” for how the bacteria interact, with an interface definition under the name “ports_schema” for the internal values of biomass, f-plasmid presence in a given cell,

and health (for ease of understanding when looking at mass sums of data). Overall, this was meant as a method for connecting a given bacteria cell to an outside lethality port for ampicillin particle - bacteria cell interactions. This also tracked the location of the cell across the “surface” of the simulated environment. The simulation was done in timeframes because of possible computer error (laginess or slowness) such that the simulation is deterministic, especially since the equations used for variables were often deterministic. This also helps with reproducibility, though that will be along a general trend because of the random nature of the simulation.

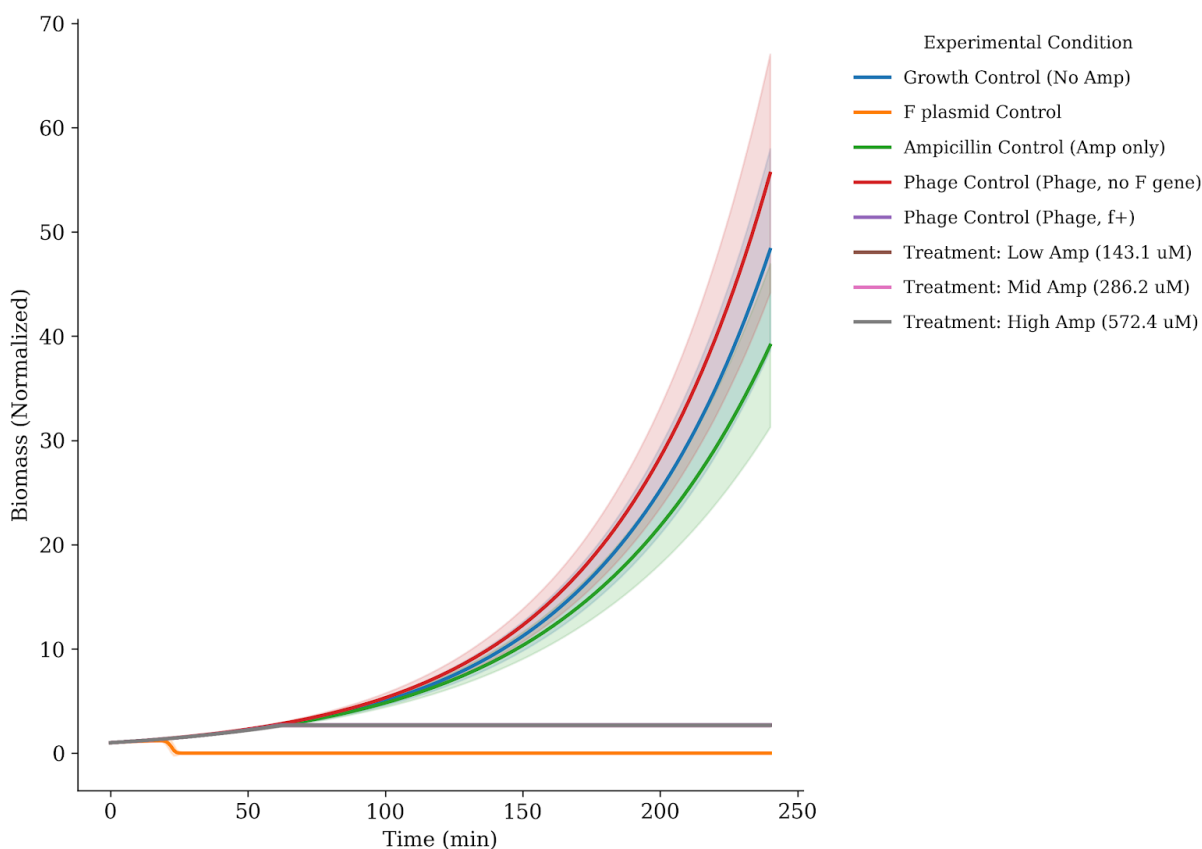
Another advantage is that deterministic values make it very convenient to find points where populations begin dying off. Furthermore, equations for Langevin fluctuations, LHM and other processes are deterministic anyway. According to Regoes et al., hill functions were used according to pharmacodynamics because the rate that bacteria is killed is nonlinear and basically ampicillin accumulates before rapidly exterminating bacteria, rather than linearly killing bacteria (Regoes et al., 2004). The first major module was GenEcoli, which was used for handling the cost of resistance and other cellular processes. The second portion of the code was used for the MS2 bacteriophage and its interaction with potential hosts. First, a first-order decay constant was applied and the phage can only infect and replicate in a given cell if the cell is expressing a F+ factor, and as the internal virion count reaches the optimal size, the cell can burst and newly created phages are released into the immediate environment. Next, the composer from Vivarium defined the topology of the system and utilized ports to link internal biologies to external fields. This means that antibiotic and phage logic only interacted through bacteria cells, resembling how the loss of a gene due to MS2 bacteriophage pressures results in ampicillin sensitivities. The final section of the code was general plotting and executing Monte Carlo simulations, which yielded

two data sets, one being aggregated, mean results and one being pure raw data. Both of these results were in .csv forms for ease of access.

Results

Figure 1

Phage-Ampicillin-Bacteria Interactions Including Controls



As seen, controls and treatments often dipped much lower than the growth and various other controls. This shows that there is a start contrast between growth controls and treatment controls, mimicking what is expected based on the literature. Very interestingly, the bacteriophage control that includes the f^+ gene did not sharply decrease to 0, meaning that the bacteriophage MS2 was not rapid enough to lyse all cells before they could shed the F plasmid.

Table 3*Mean and Standard Deviation of Biological Metrics for Ampicillin Control (Amp only)*

Time (minutes)	Health M (SD)	Biomass M (SD)	Virion M (SD)	F-Plasmid M (SD)
0	100.00 (0.00)	1.00 (0.00)	0.00 (0.00)	1.00 (0.00)
60	97.27 (0.15)	2.59 (0.13)	0.00 (0.00)	1.00 (0.00)
120	94.54 (0.29)	6.56 (0.65)	0.00 (0.00)	1.00 (0.00)
180	91.82 (0.44)	16.21 (2.42)	0.00 (0.00)	1.00 (0.00)
240	89.09 (0.58)	39.12 (7.83)	0.00 (0.00)	1.00 (0.00)

Table 4*Mean and Standard Deviation for Treatment: High Amp (572.4 μ M)*

Time (minutes)	Health M (SD)	Biomass M (SD)	Virion M (SD)	F-Plasmid M (SD)
0	100.00 (0.00)	1.00 (0.00)	0.00 (0.00)	1.00 (0.00)
60	97.21 (0.15)	2.59 (0.13)	1493 (0.00)	1.00 (0.00)
120	-0.05 (0.00)	2.67 (0.14)	1517.93 (0.00)	1.00 (0.00)

180	-0.05 (0.00)	2.67 (0.14)	1517.93 (0.00)	1.00 (0.00)
240	-0.05 (0.00)	2.67 (0.14)	1517.93 (0.00)	1.00 (0.00)

In comparison, these results show that ampicillin treatments clearly succeed in trial and all bacteria die out around 2 hours. However, one issue is that bacteriophage nor were not considered ampicillin resistance were considered within the simulation and it was assumed that genes would be limited to the F plasmid because most bacteria would die by the time any bacteria resistance genes would appear. Tuomanen et al. shows that ampicillin induced lysis is extremely rapid and happens around 60 to 120 minutes, and Jain et al. showed that death of populations often occurred before populations could actually adapt to bacteriophages (Tuomanen et al., 1986; Jain et al., 2009). But, towards numerical analysis, factorial ANOVA demonstrates clear synergy yet the Bliss Independence model, which assumes independent interactions, yields slight negativity which can be attributed to slight computational differences, and is just a lack of synergy.

Table 5

Factorial ANOVA for Biomass Values at the Final Time Point

Source	<i>SS</i>	<i>dF</i>	<i>F</i>	<i>p</i>
Phage	168,264.15	1	4,307.57	< .001
Ampicillin	2,125.41	1	54.41	< .001
Phage + Ampicillin	2,089.42	1	53.49	< .001

Error (Residual)	15,468.72	396	N/A	N/A
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Note: SS = Sum of Squares, dF = Degrees of Freedom, total N=400 [across 4 variations of 100 trials, not every type of trial was considered for factorial ANOVA based off of mathematical requirements].

The low p values demonstrate statistical significance, and large residual errors are a reflection of large amounts of values, where small differences become very large values, in reality the error is much smaller, around 10.8770%.

Table 6

Bliss Independence Synergy Analysis

Metric	Value
Observed Effect (E_{obs})	0.9447
Expected Effect (E_{exp})	0.9545
Synergy Score ($E_{\text{obs}} - E_{\text{exp}}$)	-0.0099

Note: expected effects are just the application of MS2 phage and ampicillin controls applied individually to the Bliss Independence Equation.

Looking at the synergy value, it is below 0 which shows slight antagonism, the MS2 bacteriophage or ampicillin hinders each other. However, because of how small the value is, it can just be attributed to standard deviation from the Monte Carlo simulations, and the null hypothesis is accepted. It seemed apparent that the MS2 bacteriophage was so dominant that the majority of bacteria died rapidly and ampicillin simply had a slight neutralizing effect and simply

killed the small population of bacteria that lost the F plasmid. This is in line with the literature, as it stated that bacteria would likely be neutralized before gaining any sort of genetic mutation that allowed for bacteriophage resistance, and it can be seen that the bacteria died before having the opportunity to release the F plasmid.

Conclusion

Gap of Knowledge

This study demonstrates that in the sense of phage-antibiotic synergisms, relying simply on evolutionary forces are not enough to apply a quicker and more rapid response towards bacteria elimination. The results indicated possibilities of phage-antibiotic resistance but did not yield any, and were not statistically significant when including both data analysis methods. This means that overall, no synergism was demonstrated and male-specific bacteriophages show no advantage over normal lysing bacteriophages, and instead are less efficient.

Implications

By demonstrating the lack of synergy, there is an increased understanding on the impacts of evolutionary pressure when targeting two different structures. However, bacteriophage cocktails and other plasmid-based bacteriophages still remain viable solutions. Multi-antibiotic resistance and other factors should be considered in future studies, in order to further examine bacteriophage-induced selective pressures against bacteria with multiple defensive conditions.

Limitations and Future Exploration

Limitations in this study are largely attributed to detail in the simulation. Possibly, with various pH values, temperature values, coded bacteria resistance mechanisms, multiple-antibiotic types and resistances, bacteriophage resistance included or even bacteriophage cocktails, different synergistic results could have been explored. Interestingly, bacteriophage resistance

islands are often included on the same gene as antibiotic resistance plasmids, which makes an interesting target for future evolutionary force studies (Picton et al., 2021). The future of F-plasmid simulation can even follow along the lines of AI, utilizing heavily detailed simulations to detect and understand the trend of resistance in populations, or even research towards artificially introducing a population with F plasmids and introducing bacteriophages to target these plasmids.

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Appendix A

Code

The code was ran in Anaconda and Jupyter Notebook.

```
import pandas as pd

import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import qmc, norm
from vivarium.core.process import Process
from vivarium.core.composer import Composer
from vivarium.core.engine import Engine

class GenEcoliProcess(Process):
    def ports_schema(self):
        return {
            'internal': {
                'biomass': {'_default': 1.0, '_updater': 'accumulate', '_emit': True},
                'f_plasmid': {'_default': 1, '_updater': 'set', '_emit': True},
                'cell_state': {'_default': 100.0, '_updater': 'accumulate', '_emit': True},
                'base_fitness': {'_default': 1.0, '_updater': 'set'}
            },
            'boundary': {
                'lethality_port': {'_default': 0.0},
                'location': {'_default': [0.5, 0.5], '_emit': True}
            }
        }

    def next_update(self, timestep, states):
        health = states['internal']['cell_state']
        biomass = states['internal']['biomass']
        f_present = states['internal']['f_plasmid']
        lhs_fitness = states['internal']['base_fitness']
        amp = states['boundary']['lethality_port']

        if health <= 0:
            return {}

        e_max = 0.05
        ec_50 = 160.0
```

```

base_s_amp = (e_max * (amp**2.0)) / (ec_50**2.0 + amp**2.0)

s_amp_noise = np.random.normal(1.0, 0.10)
s_amp = max(0.0, base_s_amp * s_amp_noise * (1.0 / lhs_fitness))

kill_coefficient = 0.01 if f_present else 1.0
death_delta = -s_amp * kill_coefficient * 100.0 * timestep

if health + death_delta <= 0:
    return {'internal': {'biomass': -biomass, 'cell_state': -health}}

base_mu = 0.01608 if f_present else 0.01667
mu_noise = np.random.normal(1.0, 0.05)
mu = max(0.001, base_mu * mu_noise * lhs_fitness)

effective_mu = mu * (max(0, health + death_delta) / 100.0)
growth_delta = biomass * (np.exp(effective_mu * timestep) - 1)

return {'internal': {'biomass': growth_delta, 'cell_state': death_delta}}

class MS2PhageProcess(Process):
    def ports_schema(self):
        return {
            'infection_port': {'_default': 0.0},
            'virion_count': {'_default': 0.0, '_updater': 'accumulate', '_emit': True},
            'f_status': {'_default': 0, '_updater': 'set', '_emit': True},
            'state': {'_default': 100.0, '_updater': 'accumulate', '_emit': True},
            'exchange': {'_default': 0.0, '_updater': 'accumulate', '_emit': True}
        }

    def next_update(self, timestep, states):
        phage_env = states['infection_port']
        f_present = states['f_status']
        health = states['state']
        current_virions = states['virion_count']

        update = {}
        if health <= 0:
            if current_virions > 0 and current_virions < 1500:
                return {'virion_count': -current_virions}

```

```

    return {}

    k_decay = 1.5e-4
    decay_amount = current_virions * (1 - np.exp(-k_decay * timestep))
    if decay_amount > 0:
        update['virion_count'] = -decay_amount

    loss_rate_per_gen = 1e-4
    gen_time = 20.0
    lambda_loss = -np.log(1 - loss_rate_per_gen) / gen_time
    prob_loss = 1 - np.exp(-lambda_loss * (1.0 + phage_env * 0.1) * timestep)

    if f_present:
        if np.random.random() < prob_loss:
            update['f_status'] = 0

        if current_virions < 1500:
            replication_rate = 0.5 * phage_env * timestep
            update['virion_count'] = update.get('virion_count', 0) + replication_rate
        else:
            update['state'] = -health
            update['exchange'] = 1500.0

    return update

class EcoliSynergyAgent(Composer):
    defaults = {'agent_id': 'ecoli_0', 'ms2': {}, 'ecoli': {}}
    def generate_processes(self, config):
        return {
            'metabolism': GenEcoliProcess(config['ecoli']),
            'ms2_crn': MS2PhageProcess(config['ms2']),
        }
    def generate_topology(self, config):
        return {
            'metabolism': {'internal': ('internal',), 'boundary': ('boundary',)},
            'ms2_crn': {
                'infection_port': ('..', '..', 'fields', 'phage'),
                'virion_count': ('internal', 'virions'),
                'f_status': ('internal', 'f_plasmid'),
                'state': ('internal', 'cell_state'),
            }
        }

```

```

        'exchange': ('boundary', 'phage_transfer')
    }
}

def run_experimental_condition(amp_concentration, initial_phage, is_f_plus, sim_time=240,
seed=None, lhs_fitness=1.0):
    if seed is not None: np.random.seed(seed)
    composer = EcoliSynergyAgent({})
    agent_bundle = composer.generate()
    full_composite = {
        'processes': {'agents': {'ecoli_0': agent_bundle['processes']}},
        'topology': {'agents': {'ecoli_0': agent_bundle['topology']}},
        'steps': {}, 'flow': {}, 'state': {}
    }
    initial_state = {
        'fields': {'ampicillin': float(amp_concentration), 'phage': float(initial_phage)},
        'agents': {
            'ecoli_0': {
                'internal': {
                    'biomass': 1.0, 'f_plasmid': 1 if is_f_plus else 0,
                    'cell_state': 100.0, 'virions': 0.0, 'base_fitness': lhs_fitness,
                    'metabolites': {'atp': 100.0}
                },
                'boundary': {
                    'location': [0.5, 0.5], 'phage_transfer': 0.0,
                    'lethality_port': float(amp_concentration)
                }
            }
        }
    }
    sim = Engine(composite=full_composite, initial_state=initial_state)
    sim.update(sim_time)
    return sim.emitter.get_data()

def run_monte_carlo(condition_params, N=100, sim_time=240):
    all_trials_data = []
    sampler = qmc.LatinHypercube(d=1, seed=42)
    sample = sampler.random(n=N)
    lhs_fitness_values = norm.ppf(sample, loc=1.0, scale=0.05).flatten()
    for i in range(N):

```

```

data = run_experimental_condition(
    amp_concentration=condition_params['amp'],
    initial_phage=condition_params['phage'],
    is_f_plus=condition_params['f_plus'],
    sim_time=sim_time, seed=i, lhs_fitness=lhs_fitness_values[i]
)
all_trials_data.append(data)
return all_trials_data

if __name__ == '__main__':
    experiments = {
        "Growth Control (No Amp)": {'amp': 0, 'phage': 0, 'f_plus': True},
        "F plasmid Control ": {'amp': 500, 'phage': 0, 'f_plus': False},
        "Ampicillin Control (Amp only)": {'amp': 500, 'phage': 0, 'f_plus': True},
        "Phage Control (Phage, no F gene)": {'amp': 0, 'phage': 50, 'f_plus': False},
        "Phage Control (Phage, f+)": {'amp': 0, 'phage': 50, 'f_plus': True},
        "Treatment: Low Amp (143.1 uM)": {'amp': 143.098365, 'phage': 50, 'f_plus': True},
        "Treatment: Mid Amp (286.2 uM)": {'amp': 286.19673, 'phage': 50, 'f_plus': True},
        "Treatment: High Amp (572.4 uM)": {'amp': 572.39346, 'phage': 50, 'f_plus': True}
    }

    N_TRIALS = 100
    sim_duration = 240
    all_results = {}
    for name, params in experiments.items():
        all_results[name] = run_monte_carlo(params, N=N_TRIALS, sim_time=sim_duration)

    fig_graphs, (ax1, ax2, ax3) = plt.subplots(1, 3, figsize=(18, 6))
    colors = ['green', 'gray', 'red', 'blue', 'cyan', 'orange', 'purple', 'black']
    table_data = []
    row_labels = []
    col_labels = ["T=60 min", "T=120 min", "T=180 min", "T=240 min"]
    milestones = [60.0, 120.0, 180.0, 240.0]
    csv_data = []
    raw_csv_data = []

    for (name, ensemble), color in zip(all_results.items(), colors):
        times = sorted(list(ensemble[0].keys()))
        mean_h, std_h = [], []
        mean_b, std_b = [], []

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mean_v, std_v = [], []
mean_f, std_f = [], []
for t in times:
    h_at_t, b_at_t, v_at_t, f_at_t = [], [], [], []
    for trial_idx, trial_data in enumerate(ensemble):
        agent_data = trial_data[t]['agents'].get('ecoli_0')
        if agent_data and 'internal' in agent_data:
            h_val = agent_data['internal']['cell_state']
            b_val = agent_data['internal']['biomass']
            v_val = agent_data['internal']['virions']
            f_val = agent_data['internal']['f_plasmid']
        else:
            h_val, b_val, v_val, f_val = 0.0, 0.0, 0.0, 0.0
        h_at_t.append(h_val)
        b_at_t.append(b_val)
        v_at_t.append(v_val)
        f_at_t.append(f_val)
        raw_csv_data.append({
            'Condition': name,
            'Trial': trial_idx,
            'Time (min)': t,
            'Health': h_val,
            'Biomass': b_val,
            'Virions': v_val,
            'F-Plasmid': f_val
        })
    mean_h.append(np.mean(h_at_t))
    std_h.append(np.std(h_at_t))
    mean_b.append(np.mean(b_at_t))
    std_b.append(np.std(b_at_t))
    mean_v.append(np.mean(v_at_t))
    std_v.append(np.std(v_at_t))
    mean_f.append(np.mean(f_at_t))
    std_f.append(np.std(f_at_t))
    csv_data.append({
        'Condition': name,
        'Time (min)': t,
        'Mean Health': np.mean(h_at_t),
        'Std Health': np.std(h_at_t),
        'Mean Biomass': np.mean(b_at_t),

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        'Std Biomass': np.std(b_at_t),
        'Mean Virions': np.mean(v_at_t),
        'Std Virions': np.std(v_at_t),
        'Mean F-Plasmid': np.mean(f_at_t),
        'Std F-Plasmid': np.std(f_at_t)
    })
    mean_h, std_h = np.array(mean_h), np.array(std_h)
    mean_b, std_b = np.array(mean_b), np.array(std_b)
    mean_v, std_v = np.array(mean_v), np.array(std_v)
    mean_f, std_f = np.array(mean_f), np.array(std_f)
    for var_name, m_arr, s_arr in [("Health", mean_h, std_h),
                                   ("Biomass", mean_b, std_b),
                                   ("Virions", mean_v, std_v),
                                   ("F-Plasmid", mean_f, std_f)]:
        row_labels.append(f'{name} ({var_name})')
        row_vals = []
        for milestone in milestones:
            idx = np.abs(np.array(times) - milestone).argmin()
            if var_name == "F-Plasmid":
                row_vals.append(f'{m_arr[idx]:.2f} ± {s_arr[idx]:.2f}')
            else:
                row_vals.append(f'{m_arr[idx]:.1f} ± {s_arr[idx]:.1f}')
        table_data.append(row_vals)
    ax1.plot(times, mean_h, label=name, color=color, linewidth=2)
    ax1.fill_between(times, np.maximum(0, mean_h - std_h), np.minimum(100, mean_h +
std_h), color=color, alpha=0.15)
    ax2.plot(times, mean_b, label=name, color=color, linewidth=2)
    ax2.fill_between(times, np.maximum(0, mean_b - std_b), mean_b + std_b, color=color,
alpha=0.15)
    ax3.plot(times, mean_v, label=name, color=color, linewidth=2)
    ax3.fill_between(times, np.maximum(0, mean_v - std_v), mean_v + std_v, color=color,
alpha=0.15)

    ax1.set_title('Mean Cell Health (0 = Dead)', fontsize=12, fontweight='bold')
    ax1.set_xlabel('Time (minutes)')
    ax1.set_ylabel('Health %')
    ax1.grid(True, linestyle='--', alpha=0.6)
    ax2.set_title('Mean Biomass Accumulation', fontsize=12, fontweight='bold')
    ax2.set_xlabel('Time (minutes)')
    ax2.set_ylabel('Biomass Units')

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ax2.grid(True, linestyle='--', alpha=0.6)
ax3.set_title('Internal Virion Count', fontsize=12, fontweight='bold')
ax3.set_xlabel('Time (minutes)')
ax3.set_ylabel('Total Virions')
ax3.grid(True, linestyle='--', alpha=0.6)

lines, labels = ax1.get_legend_handles_labels()
fig_graphs.legend(lines, labels, loc='lower center', bbox_to_anchor=(0.5, -0.05), ncol=4)
fig_graphs.tight_layout(rect=[0, 0.05, 1, 1])
plt.savefig('monte_carlo_graphs.png', dpi=300, bbox_inches='tight')

fig_table, ax_table = plt.subplots(figsize=(14, 10))
ax_table.axis('tight')
ax_table.axis('off')
mpl_table = ax_table.table(
    cellText=table_data,
    rowLabels=row_labels,
    colLabels=col_labels,
    loc='center',
    cellLoc='center'
)
mpl_table.auto_set_font_size(False)
mpl_table.set_fontsize(9)
mpl_table.scale(1.2, 1.3)
plt.title('Monte Carlo Output: Mean  $\pm$  Std Dev Over Time', fontsize=14, fontweight='bold',
pad=20)
plt.savefig('monte_carlo_table.png', dpi=300, bbox_inches='tight')

pd.DataFrame(csv_data).to_csv('monte_carlo_full_data.csv', index=False)
pd.DataFrame(raw_csv_data).to_csv('monte_carlo_raw_data.csv', index=False)
plt.show()

```